EduPulse SMT IP Asset Inventory

Purpose

This document identifies the main intellectual-property and confidential-know-how assets visible from the EduPulse SMT repository and operating model. It is intended to support legal review, incorporation, internal ownership clarity, and confidentiality drafting.

This is not a patentability opinion or legal conclusion. It is a technical asset inventory for counsel.

Business Name

EduPulse SMT

Asset Categories

The most relevant IP categories here are:

1. Copyright in source code, documentation, designs, and templates
2. Trade secrets and confidential know-how in system workflows and controls
3. Brand assets and naming
4. Contractual protection of methods, implementation detail, and operating playbooks

Core Asset Inventory

Asset	Description	Likely Protection Type	Why It Matters
Multi-tenant school architecture	Tenant-specific school scoping, role controls, and operational separation	Trade secret, copyright, confidential know-how	Core system structure for serving many schools in one platform
Role-gated workflow engine	Proprietor, headmaster, teacher, finance, admin workflow separation	Trade secret, confidential know-how	Encodes governance logic and internal-control design
Paystack fee-route architecture	School-specific payment routing by fee type using Paystack route/subaccount metadata	Trade secret, confidential know-how, copyright in code	Differentiates fee collection and payment classification
Reconciliation recovery logic	Recovery of successful gateway payments not fully recorded locally	Trade secret, confidential know-how	Protects revenue integrity and operational reliability
Tamper-evident finance ledger	HMAC-chained ledger entries and verification logic	Trade secret, confidential know-how, copyright in implementation	Strong compliance and audit differentiator

Asset	Description	Likely Protection Type	Why It Matters
Application-layer encrypted fields	ORM-backed encrypted storage of sensitive values	Copyright in implementation, confidential security know-how	Protects personal and banking data
Subscription and metering logic	Plan selection, plan state, overage, trial, downgrade handling	Trade secret, copyright, confidential know-how	Commercial model embedded in code
Academic approval workflows	Report staging, approval, and governance loops	Trade secret, copyright, confidential know-how	Part of the control-layer value proposition
Operations documentation system	SOPs, launch packets, handoff docs, incident templates, onboarding playbooks	Copyright, confidential business know-how	Turns software into a repeatable operating business
Brand and product naming	EduPulse SMT name and presentation materials	Trademark/brand, copyright in materials	Customer and legal identity

High-Value Trade Secret Candidates

The strongest trade-secret candidates appear to be:

1. The school-by-school Paystack route architecture by fee type
2. The routing-mismatch rule where the actual route used can override a reference hint
3. The reconciliation sweep and missing-payment recovery design
4. The dual use of encrypted fields plus tamper-evident finance ledgering
5. The role-gated separation between recording, approval, audit review, and proprietor oversight
6. The full operations layer that connects onboarding, billing, launch, support, incidents, and renewal management

These are more defensible as confidential know-how than individual generic features like attendance or grading screens.

Repository Evidence Points

The following files are strong evidence points for counsel:

- `backend/routes.py` Payment initialization, Paystack webhook handling, route creation, reconciliation, and control logic.
- `backend/api/models.py` Paystack route-account model, ledger-entry model, and encrypted data fields.
- `backend/security/crypto.py` Application-layer encryption implementation.
- `backend/paystack_integration.py` Subscription payment initialization, verification, and charge flows.
- `AUDIT_INTEGRITY_SYSTEM.md` Finance-ledger integrity model.

- `SECURITY.md` Security posture and deployment controls.
- `docs/operations/internal/STANDARD_OPERATING_PROCEDURES.md` Formal business operations layer.
- `docs/operations/internal/LAWYER_TECHNICAL_SYSTEM_SUMMARY.md` Lawyer-facing system description.

Ownership And Contract Recommendations

Counsel should consider ensuring the following are clearly covered in contracts:

1. All source code, system architecture, documentation, templates, and operating materials belong to EduPulse SMT or its founder entity
2. Employees and contractors assign inventions, improvements, and derivative work to the business
3. Payment-routing logic, reconciliation logic, ledger logic, pricing logic, and implementation playbooks are expressly confidential
4. Partners and schools receive limited use rights only, not ownership of the platform or underlying methods
5. Reverse engineering, workflow copying, and derivative commercial use are expressly prohibited where legally appropriate

Brand And Documentation Assets

The following should also be treated as protectable business assets:

1. The EduPulse SMT name and presentation identity
2. Launch packet materials and rollout documentation
3. School implementation guides and onboarding templates
4. Incident and support communication frameworks

These may be less technical than code, but they still reflect business methods and authored material that should remain controlled.

Practical Legal Questions

Counsel should review:

1. Whether trademark registration or name protection should be pursued for EduPulse SMT
2. Whether all contributors have signed enforceable IP assignment and confidentiality terms
3. Whether the terms of service and school contracts adequately reserve all platform IP rights
4. Whether the most valuable workflow logic should be protected primarily as trade secret rather than publicly described in excessive detail

Short Position

The most valuable legal assets in EduPulse SMT are not only the code files themselves, but the combined business method embodied in:

1. Multi-tenant school governance logic
2. Paystack routing and reconciliation control
3. Tamper-evident finance ledgering

4. Security and data-protection implementation choices
5. Operations playbooks that make the platform repeatable as a business